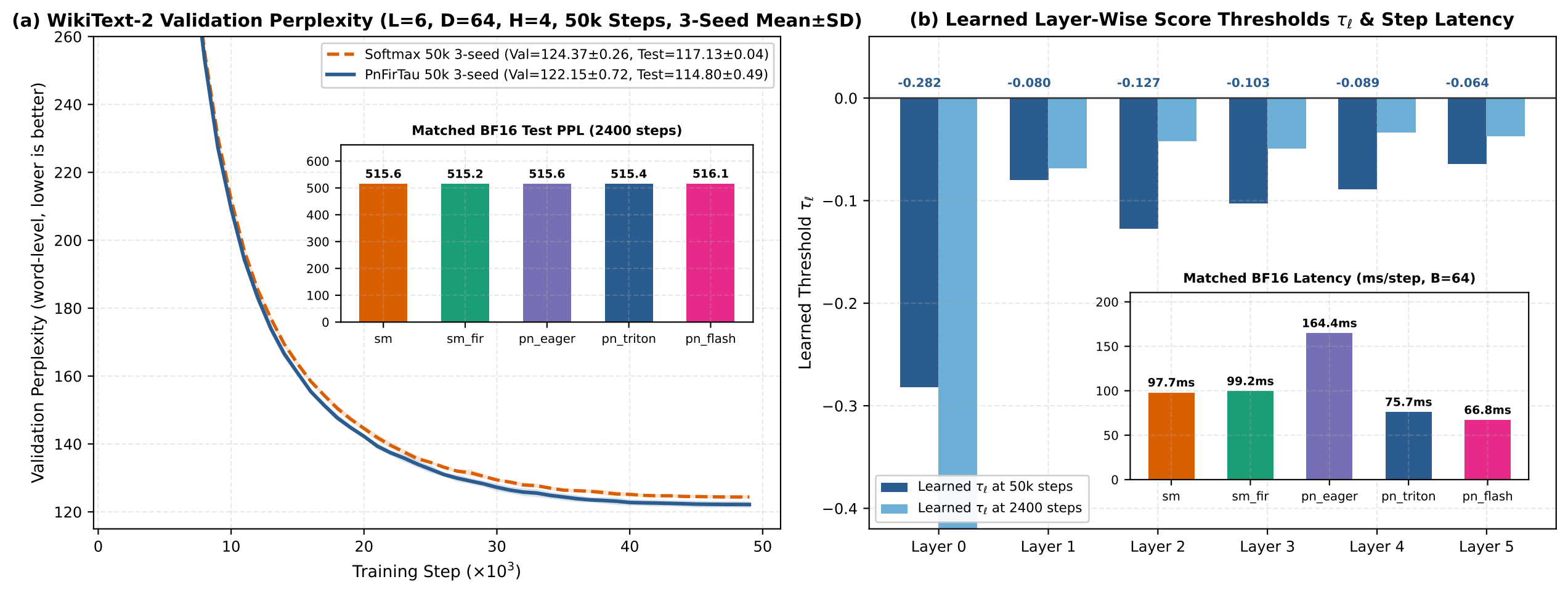
(a) WikiText-2 Validation Perplexity (L=6, D=64, H=4, 50k Steps, 3-Seed Mean±SD)
Softmax 50k 3-seed (Val=124.37±0.26, Test=117.13±0.04)
PnFirTau 50k 3-seed (Val=122.15±0.72, Test=114.80±0.49)
Validation Perplexity (word-level, lower is better)
Training Step (×10^3)
Matched BF16 Test PPL (2400 steps)
515.6
515.2
515.6
515.4
516.1
sm
sm_fir
pn_eager
pn_triton
pn_flash
(b) Learned Layer-Wise Score Thresholds τ_ℓ & Step Latency
Learned Threshold τ_ℓ
-0.282
-0.080
-0.127
-0.103
-0.089
-0.064
Learned τ_ℓ at 50k steps
Learned τ_ℓ at 2400 steps
Layer 0
Layer 1
Layer 2
Layer 3
Layer 4
Layer 5
Matched BF16 Latency (ms/step, B=64)
97.7ms
99.2ms
164.4ms
75.7ms
66.8ms